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Thermodynamic optimization of the Co–Zr system

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ABSTRACT

The constitutional and thermochemical information of the Co–Zr system is modelled using the Calphad method to obtain a reliable thermodynamic description of the system. Appropriate sublattice models are chosen to describe the Gibbs energy of all stable phases of the system. The Gibbs energies of the disordered BCC_A2 and the ordered CoZr (BCC_B2) are coupled using the order–disorder model within the framework of sublattice formalism. Results from ab initio calculations are used to aid the optimization of Gibbs energy descriptions of $CoZr_3$, $CoZr_2$, and BCC_B2 phases. Calculated phase diagram and thermochemical data are compared with the experimental data.

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1. Introduction

Knowledge of phase diagram and thermochemical properties of the Co–Zr system is very important since these alloying elements are used in several materials of technological importance. For example, Co and Zr are part of many metallic glass-forming systems like Zr–Co–Al [1] and Fe–Co–Zr–Mo–W–B [2]. These alloying elements are also present in certain superalloys [3] and high-melting carbides [4]. The metastable phase Co_5Zr is a hard magnetic phase [5], whose properties are yet to be fully studied. Hence, a complete thermodynamic description of the Co–Zr system would prove invaluable to develop alloys for new applications and also provide better understanding of the implications of these alloying elements in existing alloys.

The Co–Zr system has 10 equilibrium phases: Liquid, FCC_A1 (αCo), $Co_{11}Zr_2$ (γ), $Co_{23}Zr_6$ (δ), Laves_C15 (ϵ), BCC_B2 (ζ), $CoZr_2$ (η), $CoZr_3$ (ω), BCC_A2 (βZr) and HCP_A3 (ϵCo, αZr). Several versions of the Co–Zr phase diagram have been reported [6–13]. Pechin et al. [6] were the first to report a complete phase diagram, followed by Bataleva et al. [8]. The former diagram is shown in Fig. 1, reproduced from [14]. In [8], the $CoZr_3$ phase is shown to be forming peritectically from liquid and βZr, whereas this phase is absent in the phase diagram proposed by [6].

Chart and Putland [10] were the first to calculate the Co–Zr phase diagram, based on constitutional data from [6–8,15,16] and some estimated entropy and enthalpy values for the intermediate phases, but without using any experimental thermochemical data. Saunders and Miodownik [11] modelled the system using limited thermochemical data and assumed all the intermediate phases to be stoichiometric. Bratberg and Jansson [12] calculated the phase diagram, but not all intermediate phases were included their work. Recently, Liu et al. [13] optimized the system using stoichiometric models for the Gibbs energy of intermediate phases, except for the Laves_C15 phase. However, they have not used all the thermochemical data available in the literature for the optimization, although their own experimental phase diagram data were employed. Hence, there is a need to reassess the system.

2. Experimental data

2.1. Crystallographic data

Crystallographic data for the equilibrium phases in the system are given in Table 1. The crystal structure of the $Co_{11}Zr_2$ phase has not yet been well established, in spite of a recent attempt [17] to do so. The $Co_{23}Zr_6$ phase, earlier [6] reported to be $CoZr_4$ phase was later established [18] as having a $Mn_{23}Th_6$-type structure. Two structure types are reported for $CoZr_3$ [19] *viz.* Re_3B and Ni_3Sn. In this study, Re_3B is taken as the prototype of $CoZr_3$ since the Ni_3Sn structure is reported only by Bataleva et al. [20]. Also, Bataleva et al. later reported [8] it as having an Re_3B-type structure.

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Table 1
Crystallographic data [19].

Phase	Composition (at.% Zr)	Pearson symbol	Strukturbericht designation	Prototype
FCC_A1 (αCo)	0 to <1	cF4	A1	Cu
HCP_A3 (εCo)	∼0	hP2	A3	Mg
$Co_{11}Zr_2$ (γ)	∼15.5	–	–	–
$Co_{23}Zr_6$ (δ)	∼20.7	cF116	$D8_a$	$Mn_{23}Th_6$
Laves_C15 (ϵ)	∼25 to 34	cF24	C15	Cu_2Mg
BCC_B2 (ζ)	∼50	cP2	B2	CsCl
$CoZr_2$ (η)	∼66.7	tI12	C16	Al_2Cu
$CoZr_3$ (ω)	∼74.2 to 75.3	oC16	$E1_a$	Re_3B
BCC_A2 (βZr)	∼95 to 100	cI2	A2	W
HCP_A3 (αZr)	∼97.5 to 100	hP2	A3	Mg

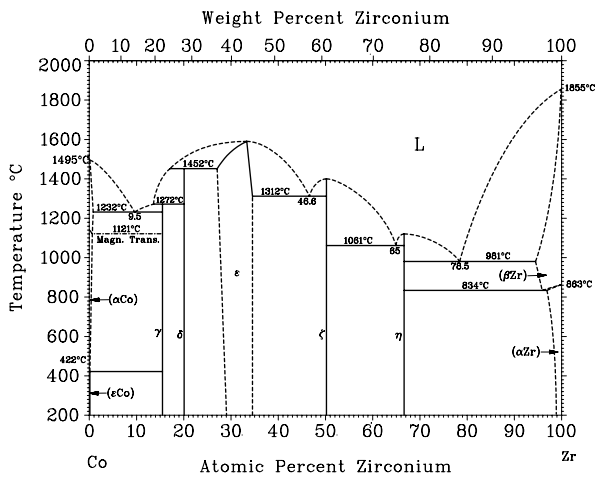

Fig. 1. Co–Zr phase diagram as reported in [14].

2.2. Constitutional data

Most comprehensive works on the phase diagram are reported by two groups [6,8]. Using metallography, X-ray diffraction, thermal analysis, and electrical resistivity measurement, Pechin et al. [6] investigated the system and determined the invariant temperatures and solidus temperatures at various compositions. Bataleva et al. [8] reported liquidus temperatures, temperatures of invariant reactions involving liquid and the homogeneity ranges of intermediate phases. They also reported the presence of a $CoZr_3$ phase which was not identified by [6]. Studies by Dobromyslov et al. [21] suggests that the $CoZr_3$ phase is metastable. According to [22], the homogeneity of Laves_C15 phase does not exceed 33.3 at.% Zr, which is slightly lower than that suggested by [8]. Recently, Liu et al. [13] used EPMA and DTA to determine the invariant reaction temperatures and phase boundaries. They have established $CoZr_3$ as a stable phase, but with a homogeneity range lower than that suggested by [8]. According to [13], the homogeneity range of Laves_C15 phase is 25–34 at.% Zr. Compared to the invariant temperatures involving liquid, congruent melting points of Laves_C15, BCC_B2, and $CoZr_2$ are imprecise. Thermal analysis data of [6] indicate that the solidus temperature for an alloy close to the ideal stoichiometry of Laves_C15 is around 1812 K. This should be close to its congruent melting point. However, in the published phase diagram it appears to be placed around 1863 K. Bataleva et al. [8] have determined this temperature to be 1833 K. Hence, in the present study, the melting point of Laves_C15 is considered to be between 1812–1833 K. Similarly, data of [6] indicate that the congruent melting point of BCC_B2 should be above 1578 K, which is the temperature of the invariant reaction Liquid ↔ Laves_C15 + BCC_B2. Melting point of $CoZr_2$ reported by [6,8] differs by about 33 K.

2.3. Thermochemical data

Experimental enthalpies of formation of all the intermediate phases, except $Co_{11}Zr_2$ and $CoZr_3$, are available in literature [23–28]. The enthalpies of the mixing of liquid in the composition range 0–30 at.% Zr at 1873 K were measured using calorimetry [29]. The calculated values of enthalpies of formation of some of the intermediate phases by Colinet et al. [30] are close to experimentally determined values. Wang et al. [31] derived an expression for the mixing enthalpy of the liquid phase by fitting a curve to the experimental data from Lück et al. [29].

The sources of experimental data used in the optimization are listed in Table 2.

3. Theoretical predictions

In addition to the experimental data, some theoretical predictions are also used in the data set for optimization. Ohodnicki et al. [32] have calculated energies of the formation of $Co_{23}Zr_6$, Laves_C15, BCC_B2, $CoZr_2$, and $CoZr_3$ from first principles. In the present work, the enthalpies of the formation of $CoZr_3$ and $Co_{11}Zr_2$ are calculated using Miedema's model [33] with improved parameters for Zr [34], since these values have not been determined experimentally. To check the reliability of the predicted enthalpy values by Miedema's method, we have also calculated the enthalpies of formation of other compounds and compared them with experimental data. The predicted values are very close to the experimental ones. In the present work, we have also used the CASTEP [35] program to perform DFT-based ab initio calculation of energy of formation of $CoZr_3$ (see Section 5). In addition to this, the excess entropies of liquid are generated using Tanaka's model [36] from the enthalpies of mixing [29]. These are used to constrain the excess entropy parameters of the liquid phase during optimization.

4. Gibbs energy models

The Gibbs energy parameters for the pure elements are taken from the SGTE compilation [37]. Liquid phase and terminal solid solutions FCC_A1 and HCP_A3 are modelled as random substitutional solutions using the Redlich–Kister [38] expression for excess Gibbs energy. The magnetic contribution to Gibbs energy [39] is included for terminal solid solutions.

The phases $Co_{11}Zr_2$ and $Co_{23}Zr_6$ are modelled as stoichiometric phases.

All non-stoichiometric intermediate phases are modelled using appropriate sublattice models [40,41], taking into account their crystallography and composition range of existence. The Laves_C15 phase is modelled using the commonly used two-sublattice model $(Co\%, Zr)_2(Co, Zr\%)_1$, with the constraints on Gibbs energies of the end-members as suggested by [42], that makes it equivalent to the Wagner–Schottky model [43].

The $CoZr_2$ phase has a Al_2Cu-type structure and hence a two-sublattice model $(Co\%, Zr)_1(Zr)_2$ is used, where Co is the major

Table 2
Experimental data from literature.

Measurement	Experimental method	Reference
Solidus	DTA	[6]
Phase boundary	XRD, metallography, Electrical resistivity measurements	[6]
Phase boundary	Metallography	[8]
Liquidus and invariant temperatures	Thermal analysis	[8]
Invariant temperature	DTA	[13]
Phase boundary	EPMA, XRD	[13]
Enthalpy of formation of intermediate phases	Direct reaction calorimetry	[24]
Enthalpy of formation of CoZr	Solution calorimetry	[25]
Enthalpy of formation of Laves phase	Solution calorimetry	[26]
Enthalpy of formation of intermediate phases	Direct reaction calorimetry	[27]
Enthalpy of formation of CoZr	Calorimetry	[28]
Enthalpy of mixing of liquid	Calorimetry	[29]

Table 3
Results from ab initio calculations.

Compound/parameters	$CoZr_3$	Zr	Co	Zr	Co	Zr
Structure	oC16	hP2	hP2 Spin polarised	tI12	cP1	cP1
Cut-off Energy (eV)	370	370	370	350	330	270
k-points used	14 × 4 × 5	17 × 17 × 8	18 × 18 × 7	11 × 11 × 11	10 × 10 × 10	8 × 8 × 8
No. of atoms in the primitive cell	8	2	2	6	1	1
Energy after geometry optimization (eV)	−9774.68043	−2562.17570	−2086.27235	−7684.74644	−1042.02599	−1280.17209

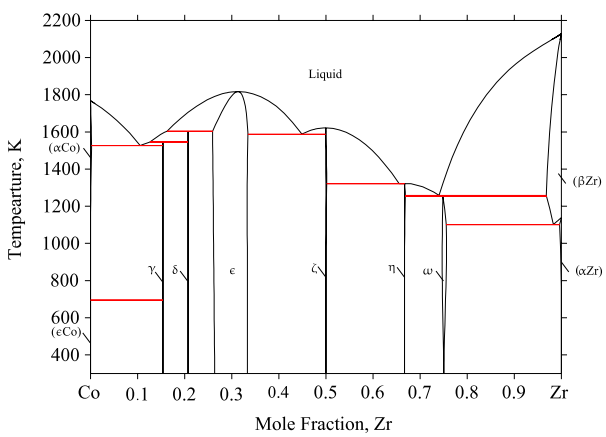

Fig. 2. The calculated Co–Zr phase diagram.

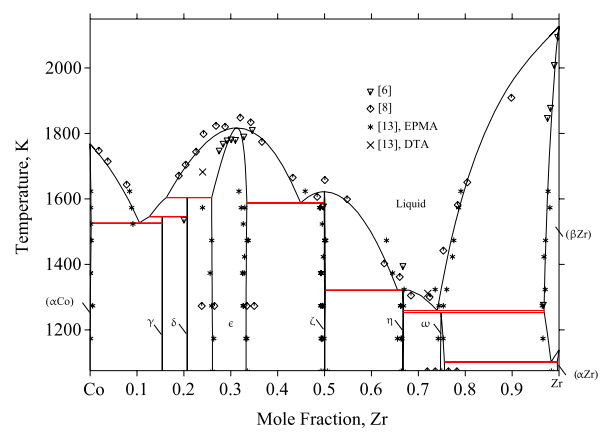

Fig. 3. Comparison of the calculated phase diagram with experimental data.

constituent in the first sublattice. Note that this type of phase is also known to have non-stoichiometry in systems such as Al–Cu, Ni–Ta.

The prototype structure of $CoZr_3$ phase is Re_3B. In a single unit cell containing 4 formula units of $CoZr_3$, 4 equivalent lattice sites are occupied by Co, another 4 by Zr and the remaining 8 by Zr. Hence, this phase is modelled using a 3-sublattice model with mixing in the first two sublattices to account for the homogeneity range. The model is given as: $(Co\%, Zr\%)_1(Co, Zr\%)_1\ (Zr)_2$, where Co is the major constituent in the first sublattice and Zr in the second sublattice. Choice of such a model is further justified since in other systems (B–Re and Fe–Zr), where such a structure type exists there is well-established non-stoichiometry for the phase.

The terminal solid solution BCC_A2 and the intermediate phase BCC_B2 are modelled using the two-sublattice order–disorder (A2–B2) model [44]. The A2 phase is modelled as $(Co, Va, Zr\%)_1(Va)_3$ and the B2 phase as $(Co\%, Va, Zr)_{0.5}(Co, Va, Zr\%)_{0.5}(Va)_3$.

5. Ab-initio calculations

For the $CoZr_3$ phase, whose structure is well established, but no experimental enthalpy data is available, we have attempted to calculate the energy of formation using the ab initio program CASTEP [35]. Perdew–Burke–Ernzerhof (PBE) GGA [45] and ultrasoft pseudopotentials are used in this work for all the ab initio calculations. An optimum cut-off energy and an optimum number of *k*-points are determined by performing single-point energy calculations, varying the two parameters one at a time. The two parameters are chosen such that the calculated energy has an accuracy of 10^{-5} eV/atom. With these two parameters fixed, the geometry optimization is carried out to obtain the energy and configuration of the compound at 0 K. Parameters for convergence control of geometry optimization are set at 5×10^{-6} eV/atom for total energy, 0.01 eV/Å for maximum force, 0.02 GPa for stress, and 5×10^{-4} Å for displacement. The geometry optimization is performed, starting with the primitive unit cell of the crystal structure. To calculate the energy of formation of the $CoZr_3$ phase, the energy of Co and Zr in their ground-state HCP structure should also be known. Hence, geometry optimization is done for Co and Zr also.

Considering the $CoZr_2$ sublattice model, there are 2 end-members namely Co:Zr and Zr:Zr. The first one corresponds to the required stable structure of the compound and the second one is a hypothetical compound. Hence, the parameter ${}^{\circ}G^{CoZr_2}_{Zr:Zr}$ corresponding to Zr:Zr is calculated. Similarly, total energies of the

Table 4
Optimized Gibbs energy parameters.

Gibbs energy parameters (in SI units)
Liquid:(Co, Zr)
${}^{0}L^{\mathrm{Liquid}}_{\mathrm{Co,Zr}} = -153\,092 + 14.484T$
${}^{1}L^{\mathrm{Liquid}}_{\mathrm{Co,Zr}} = -12\,768$
${}^{2}L^{\mathrm{Liquid}}_{\mathrm{Co,Zr}} = +34\,316 - 5.594T$
HCP_A3:(Co, Zr)$_1$(Va)$_{0.5}$
$TC^{\mathrm{HCP_A3}}_{\mathrm{Co:Va}} = +1396$
$\beta^{\mathrm{HCP_A3}}_{\mathrm{Co:Va}} = +1.35$
${}^{0}L^{\mathrm{HCP_A3}}_{\mathrm{Co,Zr:Va}} = -47\,428$
${}^{1}L^{\mathrm{HCP_A3}}_{\mathrm{Co,Zr:Va}} = -5961$
FCC_A1:(Co, Zr)$_1$(Va)$_1$
$TC^{\mathrm{FCC_A1}}_{\mathrm{Co:Va}} = +1396$
$\beta^{\mathrm{FCC_A1}}_{\mathrm{Co:Va}} = +1.35$
${}^{0}L^{\mathrm{FCC_A1}}_{\mathrm{Co,Zr:Va}} = -60\,000$
${}^{1}L^{\mathrm{FCC_A1}}_{\mathrm{Co,Zr:Va}} = -4000$
$Co_{11}Zr_2$:(Co)$_{11}$(Zr)$_2$
${}^{\circ}G^{\mathrm{Co_{11}Zr_2}}_{\mathrm{Co:Zr}} = +11 * \mathrm{GHSER_{Co}} + 2 * \mathrm{GHSER_{Zr}} + 13 * (-23\,056 + 1.638T)$
$Co_{23}Zr_6$:(Co)$_{23}$(Zr)$_6$
${}^{\circ}G^{\mathrm{Co_{23}Zr_6}}_{\mathrm{Co:Zr}} = +23 * \mathrm{GHSER_{Co}} + 6 * \mathrm{GHSER_{Zr}} + 29 * (-28\,399 + 0.914T)$
Laves_C15:(Co, Zr)$_2$(Co, Zr)$_1$
${}^{\circ}G^{\mathrm{Laves_C15}}_{\mathrm{Co:Co}} = +3 * \mathrm{GHSER_{Co}} + 15\,000$
${}^{\circ}G^{\mathrm{Laves_C15}}_{\mathrm{Zr:Co}} = +\mathrm{GHSER_{Co}} + 2 * \mathrm{GHSER_{Zr}} - 3 * (-39\,475 + 0.785T) + 30\,000$
${}^{\circ}G^{\mathrm{Laves_C15}}_{\mathrm{Co:Zr}} = +2 * \mathrm{GHSER_{Co}} + \mathrm{GHSER_{Zr}} + 3 * (-39\,475 + 0.785T)$
${}^{\circ}G^{\mathrm{Laves_C15}}_{\mathrm{Zr:Zr}} = +3 * \mathrm{GHSER_{Zr}} + 15\,000$
${}^{0}L^{\mathrm{Laves_C15}}_{\mathrm{Co:Co,Zr}} = -49\,863 + 26.539T$
${}^{1}L^{\mathrm{Laves_C15}}_{\mathrm{Co:Co,Zr}} = +41\,730$
$CoZr_2$:(Co, Zr)$_1$(Zr)$_2$
${}^{\circ}G^{\mathrm{CoZr_2}}_{\mathrm{Co:Zr}} = +2 * \mathrm{GHSER_{Zr}} + \mathrm{GHSER_{Co}} + 3 * (-30\,405 + 0.323T)$
${}^{\circ}G^{\mathrm{CoZr_2}}_{\mathrm{Zr:Zr}} = +3 * \mathrm{GHSER_{Zr}} + 3 * (+29\,000)$
${}^{0}L^{\mathrm{CoZr_2}}_{\mathrm{Co,Zr:Zr}} = -35\,000$
$CoZr_3$:(Co, Zr)$_1$(Co, Zr)$_1$(Zr)$_2$
${}^{\circ}G^{\mathrm{CoZr_3}}_{\mathrm{Co:Co:Zr}} = +2 * \mathrm{GHSER_{Co}} + 2 * \mathrm{GHSER_{Zr}} + 4 * (-40\,135 + 1.503T)$
${}^{\circ}G^{\mathrm{CoZr_3}}_{\mathrm{Zr:Co:Zr}} = +\mathrm{GHSER_{Co}} + 3 * \mathrm{GHSER_{Zr}} + 4 * (-24\,736 + 1.820T)$
${}^{\circ}G^{\mathrm{CoZr_3}}_{\mathrm{Co:Zr:Zr}} = +\mathrm{GHSER_{Co}} + 3 * \mathrm{GHSER_{Zr}} + 4 * (-24\,736 + 1.820T)$
${}^{\circ}G^{\mathrm{CoZr_3}}_{\mathrm{Zr:Zr:Zr}} = +4 * \mathrm{GHSER_{Zr}} + 4 * (+4000 + 3.50T)$
${}^{0}L^{\mathrm{CoZr_3}}_{\mathrm{Co:Co,Zr:Zr}} = +8166$
${}^{0}L^{\mathrm{CoZr_3}}_{\mathrm{Co,Zr:Co:Zr}} = +8166$
${}^{0}L^{\mathrm{CoZr_3}}_{\mathrm{Zr:Co,Zr:Zr}} = -2214$
${}^{0}L^{\mathrm{CoZr_3}}_{\mathrm{Co,Zr:Zr:Zr}} = -2214$
BCC_A2:(Co, Va, Zr)$_1$(Va)$_3$
$TC^{\mathrm{BCC_A2}}_{\mathrm{Co:Va}} = +1450$
$\beta^{\mathrm{BCC_A2}}_{\mathrm{Co:Va}} = +1.35$
${}^{\circ}G^{\mathrm{BCC_A2}}_{\mathrm{Va:Va}} = +30T$
${}^{0}L^{\mathrm{BCC_A2}}_{\mathrm{Co,Zr:Va}} = -55\,065$
${}^{1}L^{\mathrm{BCC_A2}}_{\mathrm{Co,Zr:Va}} = +3500$
${}^{0}L^{\mathrm{BCC_A2}}_{\mathrm{Co,Va:Va}} = +165\,000$
${}^{0}L^{\mathrm{BCC_A2}}_{\mathrm{Va,Zr:Va}} = +320\,000$
BCC_B2:(Co, Va, Zr)$_{0.5}$(Co, Va, Zr)$_{0.5}$(Va)$_3$
${}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Co:Co:Va}} = +0.0$
${}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Va:Va:Va}} = +0.0$
${}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Zr:Zr:Va}} = +0.0$

Table 4 (*continued*)

Gibbs energy parameters (in SI units)
${}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Co:Va:Va}} = {}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Va:Co:Va}} = (+\mathrm{GHSER_{Co}} + 107\,100) * 0.5$
${}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Zr:Va:Va}} = {}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Va:Zr:Va}} = (+\mathrm{GHSER_{Zr}} + 88\,400) * 0.5$
${}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Co:Zr:Va}} = {}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Zr:Co:Va}} = +6112 + 7.137T$
${}^{0}L^{\mathrm{BCC_B2}}_{\mathrm{Co:Co,Zr:Va}} = {}^{0}L^{\mathrm{BCC_B2}}_{\mathrm{Co,Zr:Co:Va}} = +49\,181$
${}^{0}L^{\mathrm{BCC_B2}}_{\mathrm{Zr:Co,Zr:Va}} = {}^{0}L^{\mathrm{BCC_B2}}_{\mathrm{Co,Zr:Zr:Va}} = +91\,491$
$TC^{\mathrm{BCC_B2}}_{\mathrm{Co:Zr:Va}} = TC^{\mathrm{BCC_B2}}_{\mathrm{Zr:Co:Va}} = -1175$

simple cubic (Pearson symbol: cP1) form of Co and Zr were also calculated. These values were used in fixing the values of Gibbs energy parameters ${}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Co:Va:Va}}$, ${}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Va:Co:Va}}$, ${}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Zr:Va:Va}}$, and ${}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Va:Zr:Va}}$.

The results obtained from ab initio calculations are indicated in Eqs. (1)–(4) and shown in Table 3.

$$\Delta E^{\mathrm{CoZr_3}} = \frac{(E(\mathrm{CoZr_3}) - 2 \cdot E(\mathrm{Co})/2 - 6 \cdot E(\mathrm{Zr})/2)}{8} = -0.23512\ \mathrm{eV/atom} = -22\,658\ \mathrm{J/mol} \approx \Delta^f H^{\mathrm{CoZr_3}} \quad (1)$$

$${}^{\circ}G^{\mathrm{CoZr_2}}_{\mathrm{Zr:Zr}} = 3 \times \mathrm{GHSER_{Zr}} + 3 \times 29\,016\ \mathrm{J/mol} \quad (2)$$

$${}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Co:Va:Va}} = {}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Va:Co:Va}} = (\mathrm{GHSER_{Co}} + 107\,100) * 0.5 \quad (3)$$

$${}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Zr:Va:Va}} = {}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Va:Zr:Va}} = (\mathrm{GHSER_{Zr}} + 88\,400) * 0.5. \quad (4)$$

6. Optimization

The optimization was carried out using the PARROT module [46] of Thermo-Calc [47]. A critical review of the experimental data was made prior to the preparation of the input file for optimization. While weighting the data, due consideration was given to experimental technique, purity of the starting materials, etc. The enthalpy of formation of $CoZr_3$ obtained from ab initio calculations (Eq. (1)) was used as an input in the optimization with an error of ±10%. The term ${}^{\circ}G^{\mathrm{CoZr_2}}_{\mathrm{Zr:Zr}}$ was fixed according to Eq. (2). In the description of BCC_A2, the parameter ${}^{\circ}G^{\mathrm{BCC_A2}}_{\mathrm{Va:Va}}$ was chosen to have a large positive value of $+30 \cdot T$ J/mol, as per the existing practice [48]. The parameter ${}^{0}L^{\mathrm{BCC_A2}}_{*,\mathrm{Va:Va}}$ was adjusted such that the melting points of the pure elements in the BCC_A2 state are same as the values obtained using the SGTE data. The parameters ${}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Co:Va:Va}}$, ${}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Va:Co:Va}}$, ${}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Zr:Va:Va}}$, and ${}^{\circ}G^{\mathrm{BCC_B2}}_{\mathrm{Va:Zr:Va}}$ were constrained according to Eqs. (3) and (4). During optimization, the liquid parameters were optimized first, followed by the terminal solid solutions and subsequently all the intermediate phases. Final optimization was done including all phases and data simultaneously, until the sum of squares of error did not show any appreciable change.

7. Results and discussion

Optimized Gibbs energy functions are listed in Table 4. The phase diagram calculated using the description is shown in Fig. 2. The comparison of calculated phase diagram with experimental data from literature is shown in Fig. 3. Considering uncertainties in the experimental data, there is reasonable agreement between the experimental and calculated phase diagrams. Enthalpies of formation of intermediate phases of the system are listed in Table 5. There is good agreement between the experimental, Miedema and ab initio values of enthalpies of formation of intermediate phases with the calculated values. In Fig. 4 the enthalpy of mixing of liquid is plotted as a function of composition. The invariant reaction temperatures are listed in Table 6.

The strength of the present work is substantially more thermochemical data used for the optimization than by [13]. Our optimization is somewhat more biased towards thermochemical

Table 5
Comparison between experimental/theoretical and optimized values of enthalpies of formation of intermediate phases (J/mol of atoms).

Phase	Temperature (K)	Experimental	Optimized	Reference	Standard deviation
$Co_{11}Zr_2$	298	−19 889	−15 846	Miedema [this work]	5000
$Co_{23}Zr_6$	298	−21 900	−21 641	ab initio [32]	2000
$Co_{23}Zr_6$	298	−26 482	–	Miedema [this work]	Not used
$Co_{23}Zr_6$	298	−32 000	–	Colinet [30]	Not used
$Co_{23}Zr_6$	1596	−29 800	−27 909	[24]	1500
Laves_C15	298	−35 800	−33 794	[27]	600
Laves_C15	298	−37 667	,,	[26]	3670
Laves_C15	1708	−41 000	−38 866	[24]	1600
Laves_C15	298	−31 680	–	ab initio [32]	Not used
Laves_C15	298	−38 166	–	Miedema [this work]	Not used
Laves_C15	298	−41 000	–	Colinet [30]	Not used
BCC_B2	298	−35 800	−36 502	[27]	700
BCC_B2	298	−41 000	,,	[28]	3000
BCC_B2	1030	−43 050	−41 144	[25]	3900
BCC_B2	1512	−42 200	−43 790	[24]	1000
BCC_B2	298	−28 830	–	ab initio [32]	Not used
BCC_B2	298	−40 135	–	Miedema [this work]	Not used
BCC_B2	298	−42 000	–	Colinet [30]	Not used
$CoZr_2$	298	−26 300	−27 564	ab initio [32]	2000
$CoZr_2$	298	−26 400	,,	[27]	1000
$CoZr_2$	1290	−33 000	−31 166	[24]	2000
$CoZr_2$	298	−29 437	–	Miedema [this work]	Not used
$CoZr_2$	298	−34 000	–	Colinet [30]	Not used
$CoZr_3$	298	−22 111	−18 899	Miedema [this work]	5000
$CoZr_3$	298	−22 440	,,	ab initio [32]	2000
$CoZr_3$	298	−22 658	,,	ab initio [this work]	2000

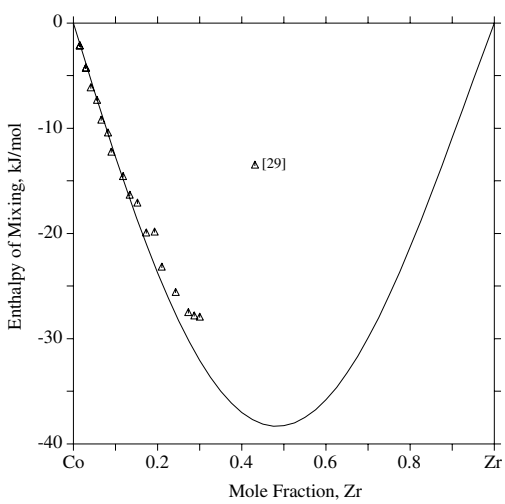

Fig. 4. Calculated and experimental values of enthalpy of mixing of liquid.

data, whereas [13] relied more on the constitutional information. This is a system where we believe that thermochemical data are relatively more reliable than the phase diagram data. During the optimization we noticed that any attempt to force fit the invariant and congruent melting temperatures lead to poor fit to the enthalpy of formation of intermediate phases. Note that purity of starting materials in most phase diagram investigation were not very good ([6]: Co 99.6%, Zr 99.94%, [8]: Co:99.9%, Zr 99.9%, [13]: Co 99.8%, Zr 99.7%).

8. Conclusions

Constitutional and thermochemical data concerning the Co–Zr system available in the literature were assessed carefully and optimized using the Calphad method with the aid of the software package Thermo-Calc. Results of ab initio calculations and Miedema's method were combined with the experimental data and used in the optimization. A reasonable agreement between the experimental and calculated phase diagram is obtained. An optimal set of parameters leading to a reliable thermodynamic description of the Co–Zr system is presented.

Table 6
Comparison between experimental and calculated invariant reaction temperatures.

Invariant reaction	Reaction temperature (K)			
	This work	Ref. [6]	Ref. [8]	Ref. [13]
Liquid ↔ Laves_C15	1817	>1812	1833	
Liquid ↔ BCC_B2	1622	>1578	1643	
Liquid ↔ $CoZr_2$	1323	1396	1363	
Liquid ↔ FCC_A1 + $Co_{11}Zr_2$	1527	1499	1513	1503
Liquid + $Co_{23}Zr_6$ ↔ $Co_{11}Zr_2$	1545	1539	1543	1544
Liquid + Laves_C15 ↔ $Co_{23}Zr_6$	1604	1721	1623	1616
Liquid ↔ Laves_C15 + BCC_B2	1588	1584	1533	1586
Liquid ↔ BCC_B2 + $CoZr_2$	1321	1334	1313	1337
BCC_A2 + $CoZr_2$ ↔ $CoZr_3$	1253			1254
Liquid ↔ $CoZr_2$ + BCC_A2	1259	1255		1258
BCC_A2 ↔ $CoZr_3$ + HCP_A3	1102			1111

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Appendix. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.calphad.2010.02.006.

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